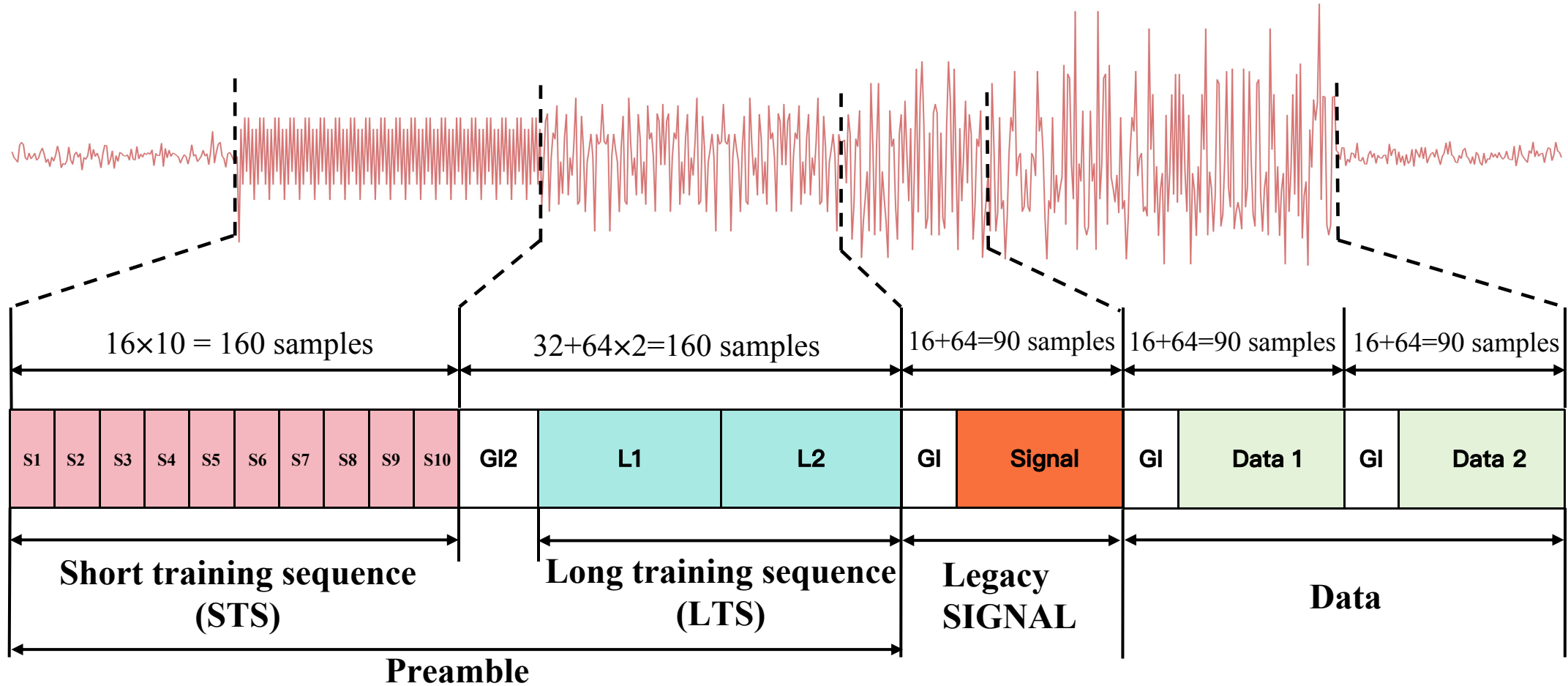
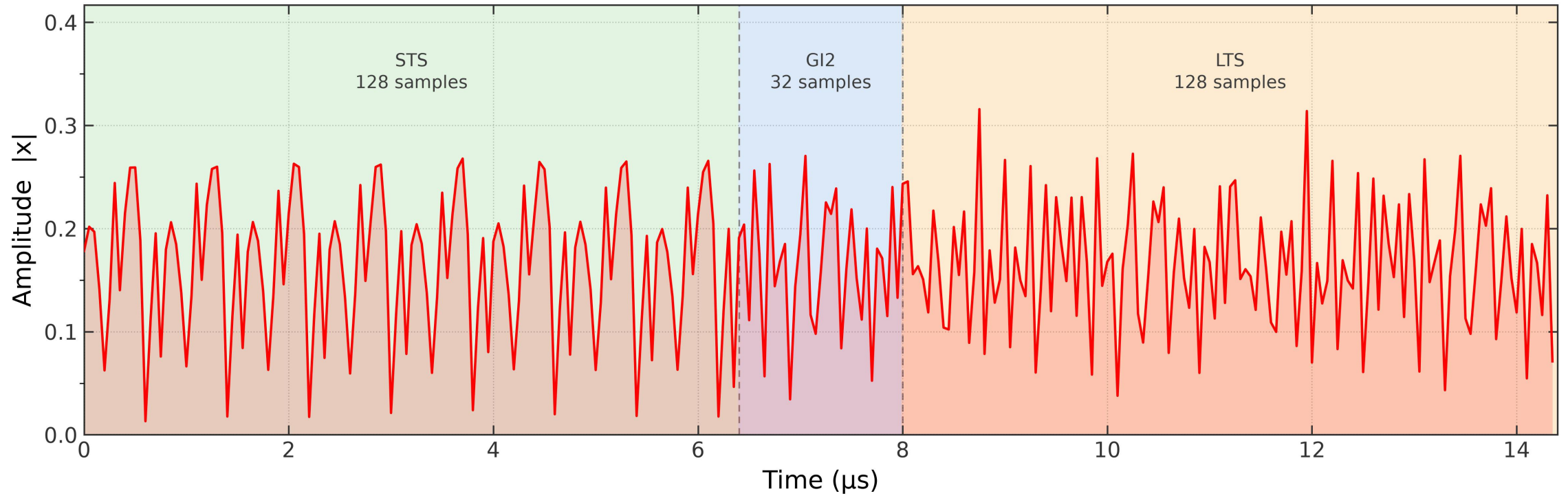


IEEE 802.11g Wi-Fi Signal

IEEE 802.11g 2.4 GHz OFDM Wi-Fi Preamble



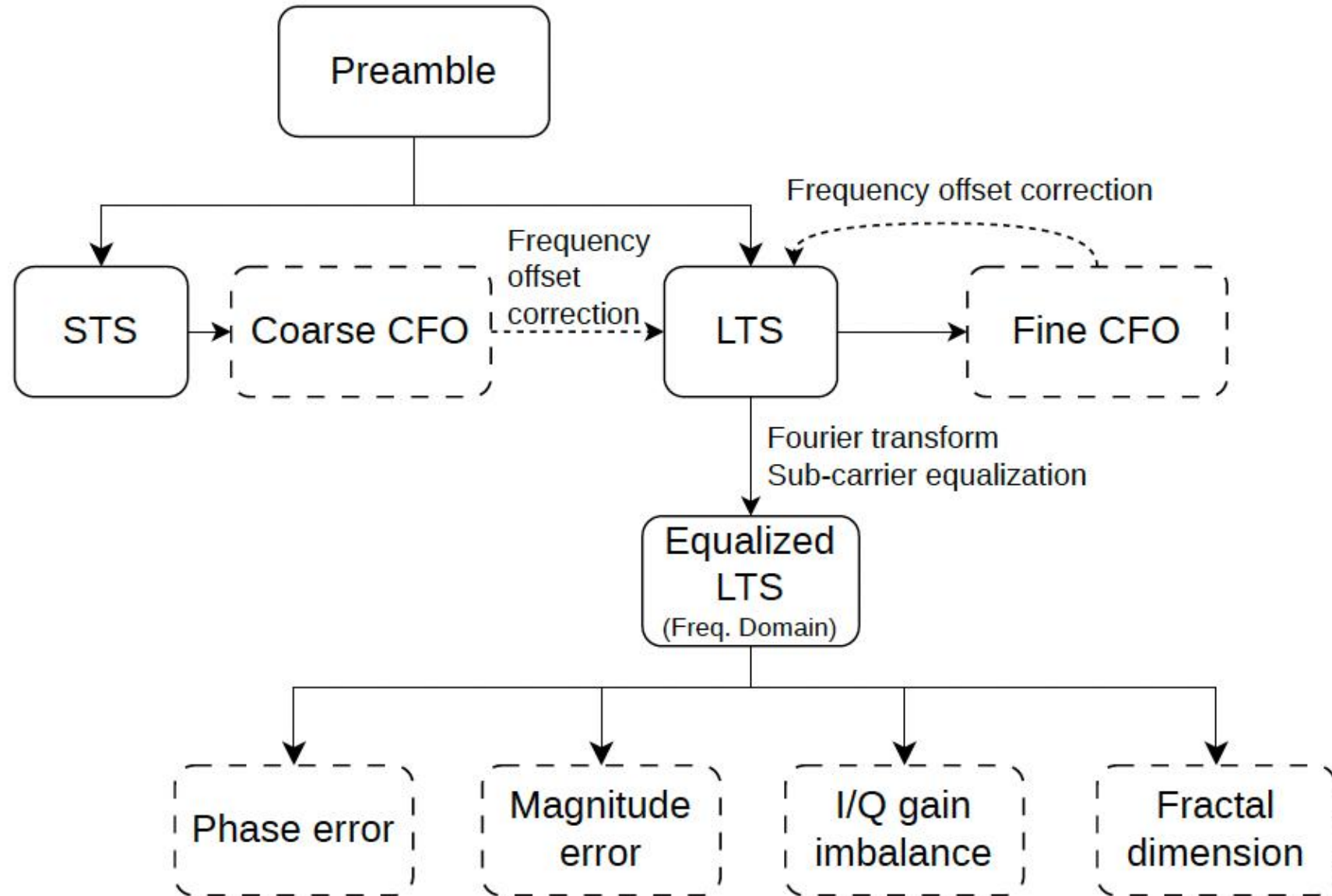
Preamble used in this tutorial



From S3 to S10, 128 samples.
S1 and S2 are dropped.

File:24_d7_eb_38_c7_e8_pre.csv
frame number:119

Feature extraction workflow



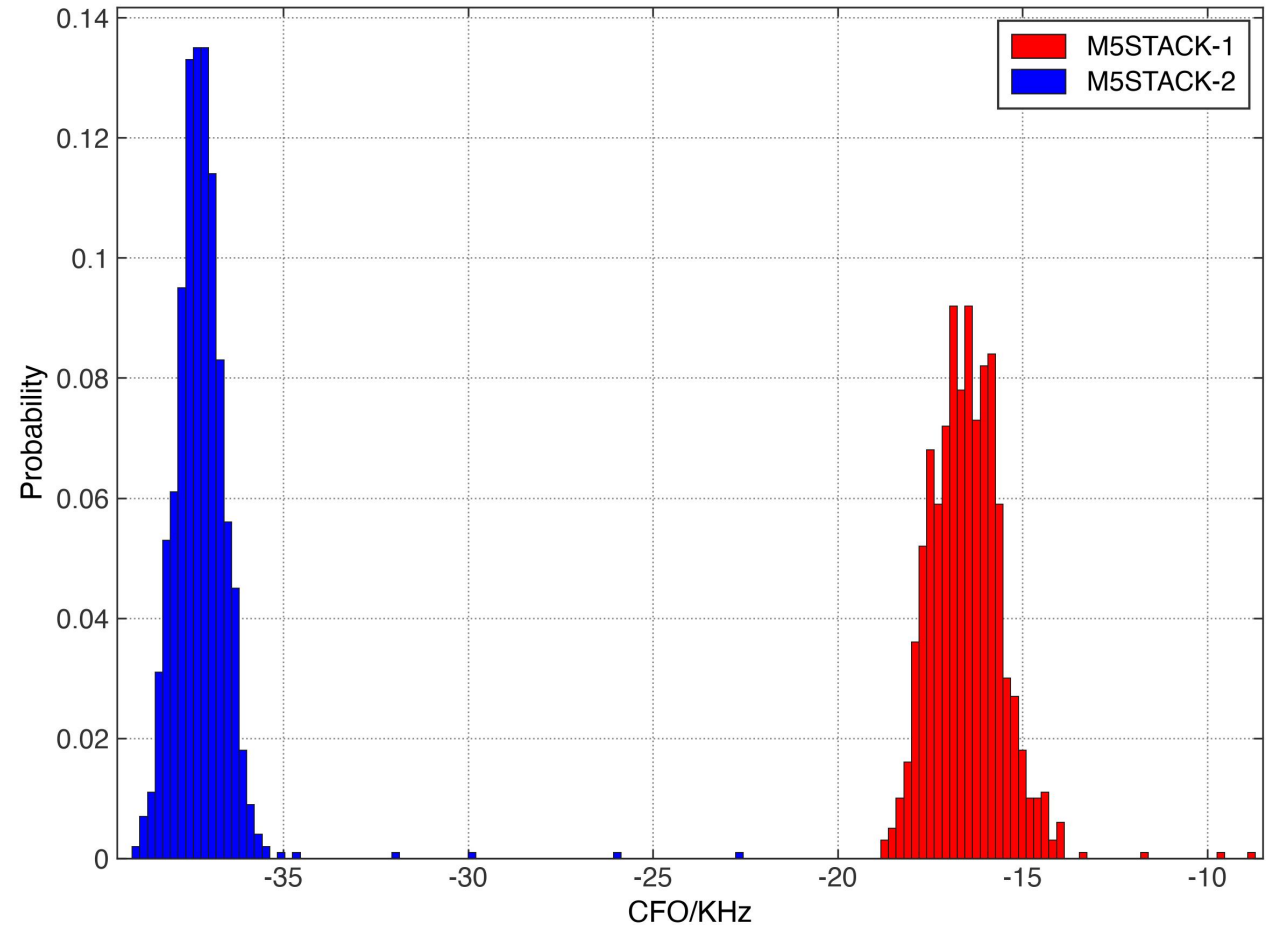
Representative RF Features - CFO

Carrier Frequency Offset (CFO)

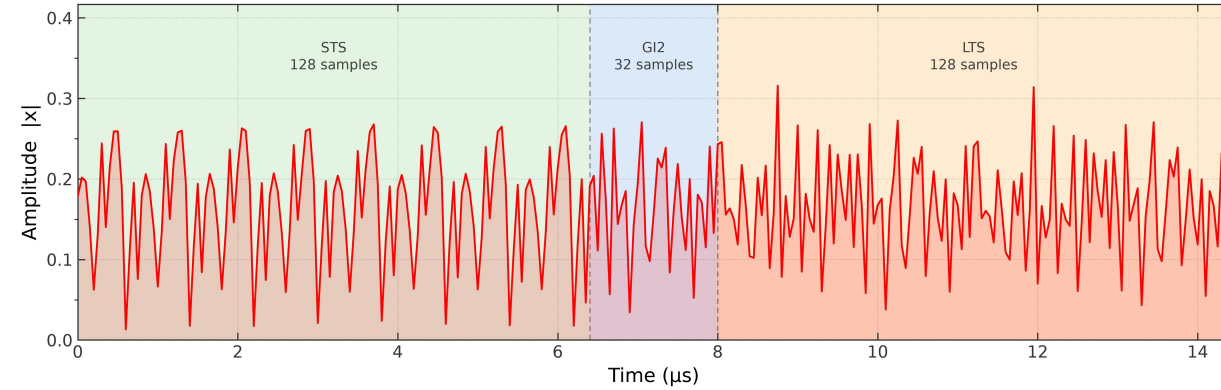
Difference between the received carrier frequency and the expected carrier frequency.

Main sources

1. Oscillator mismatch
2. Doppler shift due to mobility



Representative RF Features - CFO



$$\mathbf{X} = \left[\underbrace{x_1, \dots, x_{128}}_{\text{STS}}, \dots, \underbrace{x_{161}, \dots, x_{288}}_{\text{LTS}} \right].$$

Use STS to calculate Coarse CFO

$$\alpha_S = \frac{1}{16} \angle \left(\sum_{i=1}^{112} \mathbf{x}[i]^\dagger \mathbf{x}[i + 16] \right).$$

Use Coarse CFO to correct LTS

$$\hat{\mathbf{x}}[j] = \mathbf{x}[160 + j] e^{-j(j-1)\alpha_S}, \quad j = 1, 2, \dots, 128.$$

$$j = \sqrt{-1}$$

Use LTS to calculate Fine CFO

$$\alpha_L = \frac{1}{64} \angle \left(\sum_{i=1}^{64} \hat{\mathbf{x}}[i]^\dagger \hat{\mathbf{x}}[i + 64] \right).$$

The total CFO can be obtained as

$$\alpha_F = \alpha_S + \alpha_L.$$

Then, the fine CFO correction can be expressed as

$$\check{\mathbf{x}}[j] = \hat{\mathbf{x}}[j] e^{-j(j-1)\alpha_L}, \quad j = 1, 2, \dots, 128.$$

Here, we focus on the LTS after coarse and fine CFO corrections.

$$\check{\mathbf{x}}_1[m], \quad \check{\mathbf{x}}_2[m], \quad m = 0, 1, \dots, 63.$$

A 64-point FFT converts each CFO-corrected LTS from the time domain to the subcarrier domain.

$$X_n[k] = \sum_{m=0}^{63} \check{x}_n[m] e^{-j2\pi km/64}, \quad n \in \{1, 2\}, \quad k = 0, \dots, 63.$$

The channel response is estimated by averaging the two received LTS symbols and comparing them with the known ideal LTS.

$$\hat{H}[k] = \frac{X_1[k] + X_2[k]}{2L_{\text{ideal}}[k]}. \quad \Rightarrow \quad \text{Channel equalization} \quad I_n[k] = \frac{X_n[k]}{\hat{H}[k]}, \quad n \in \{1, 2\}.$$

After compensating for the channel effect, we further extract the following RF features.

$$I_n[k] \rightarrow \begin{cases} \text{Phase Error} \\ \text{Magnitude Error} \\ \text{I/Q gain imbalance} \\ \text{Fractal Dimension} \end{cases}$$

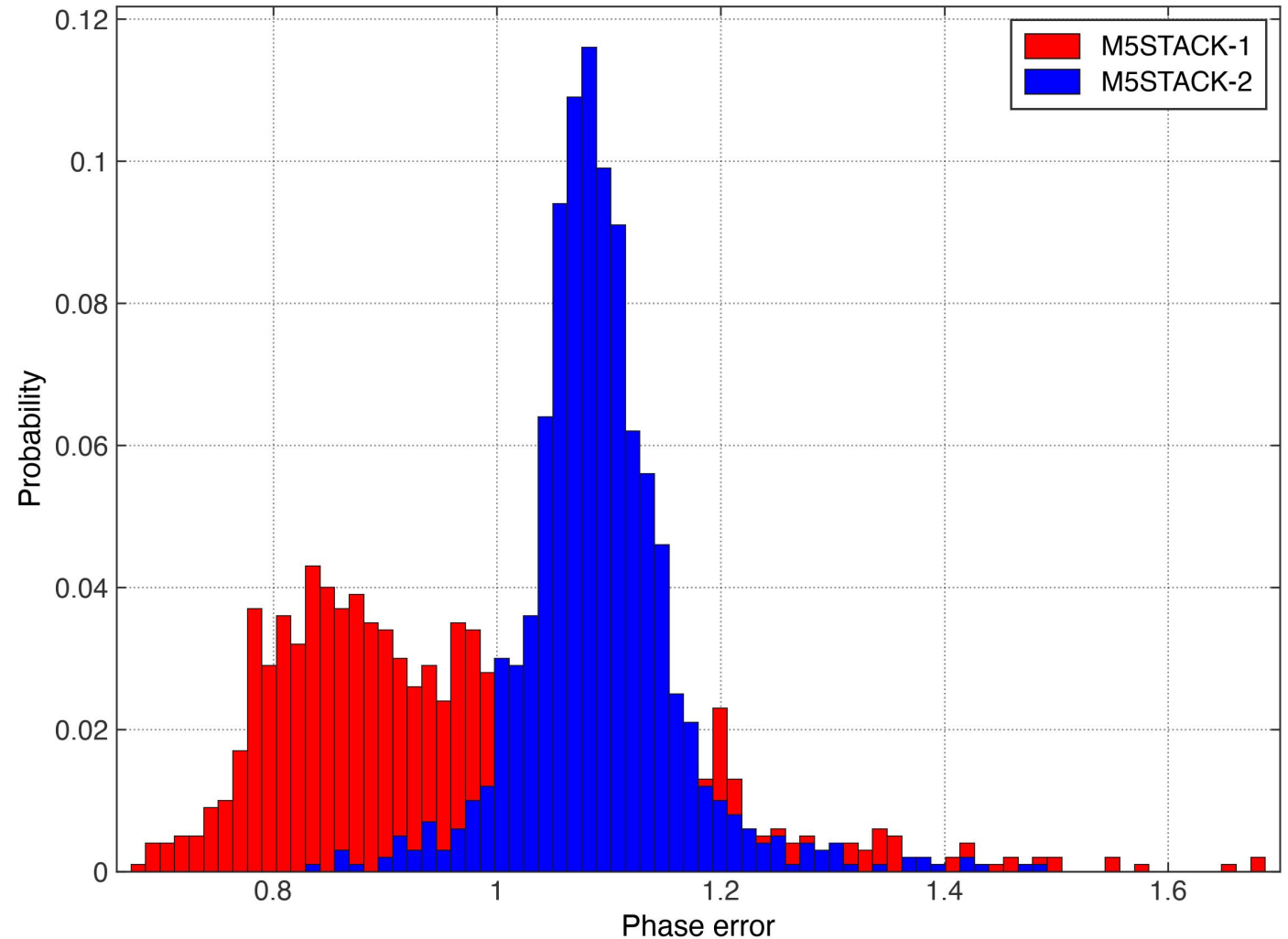
Representative RF Features – Phase Error

Phase error

Phase difference between the equalized and reference LTS

Main sources

- Residual CFO
- Oscillator phase noise
- Synchronization errors
- Other RF impairments.



$$\mathbf{p}_n[k] = \angle \mathbf{l}_n[k] - \angle \mathbf{l}_{\text{ideal}}[k], n \in \{1, 2\}, k = 1, 2, \dots, 64$$

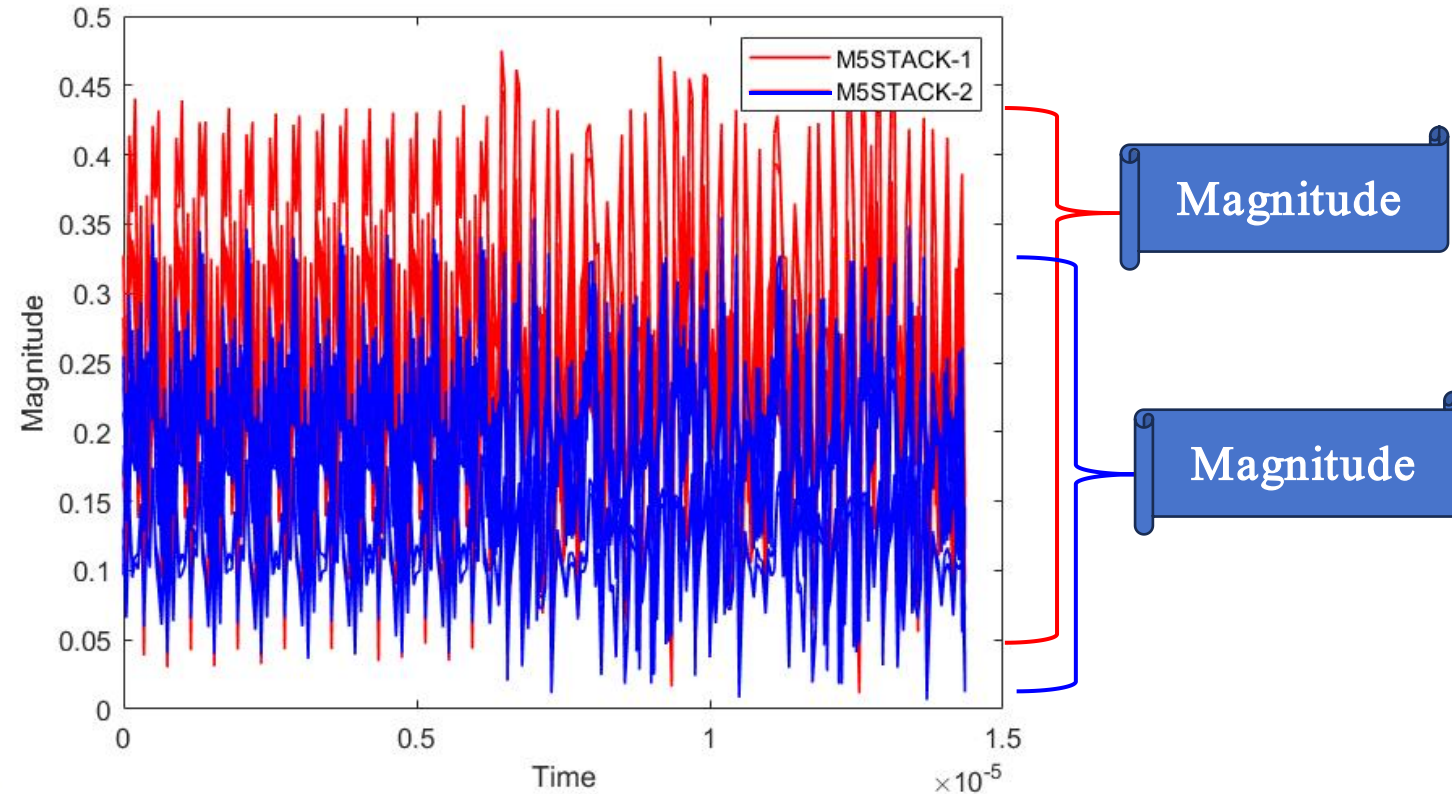
Representative RF Features – Magnitude Error

Magnitude error

Magnitude difference between the equalized and reference LTS

Main sources

- Power-amplifier characteristics
- RF front-end imperfections.



$$\mathbf{m}_n[k] = |\mathbf{l}_n[k]| - |\mathbf{l}_{\text{ideal}}[k]|, n \in \{1, 2\}, k = 1, 2, \dots, 64$$

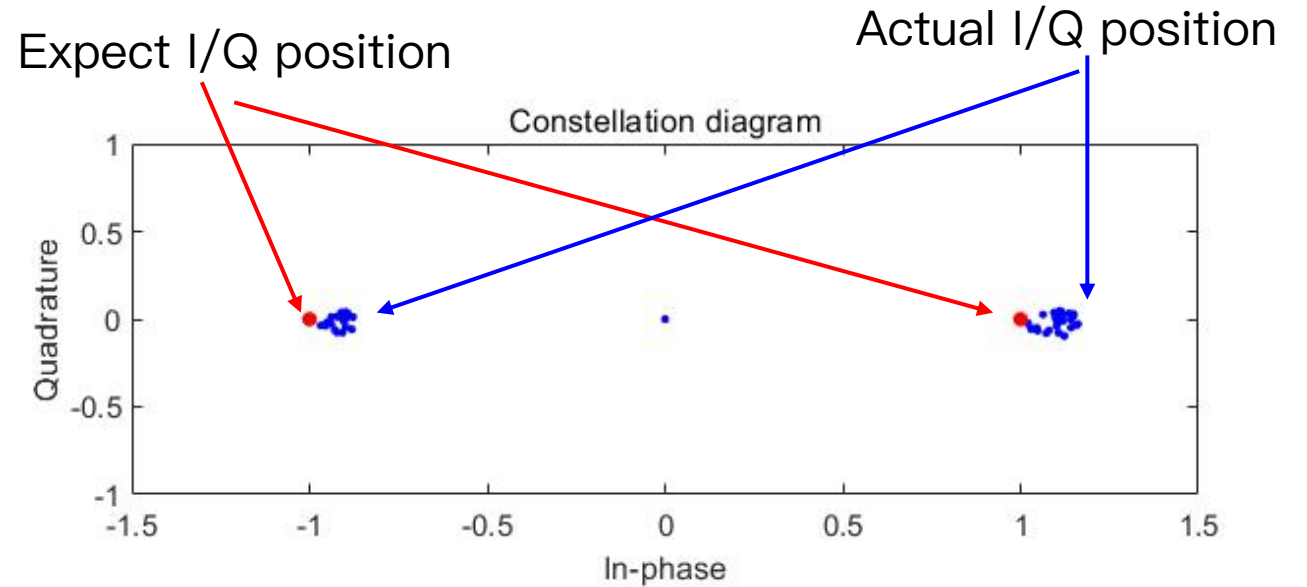
Representative RF Features - IQI

I/Q gain imbalance

Energy imbalance between the I and Q components

$$IQI_n = \frac{\|\Re\{\mathbf{I}_n\}\|^2}{\|\Im\{\mathbf{I}_n\}\|^2}, n \in \{1, 2\}$$

- Ideal balance: $IQI \approx 1$



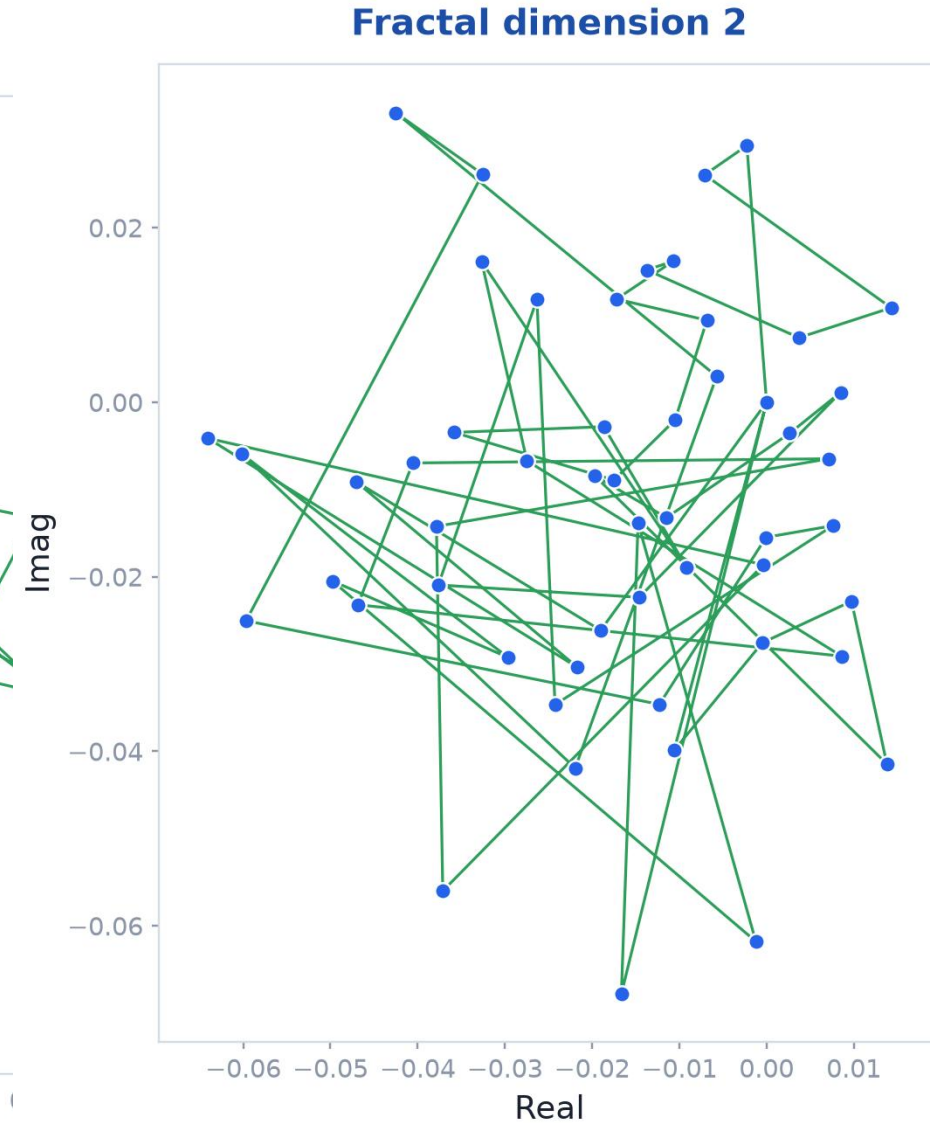
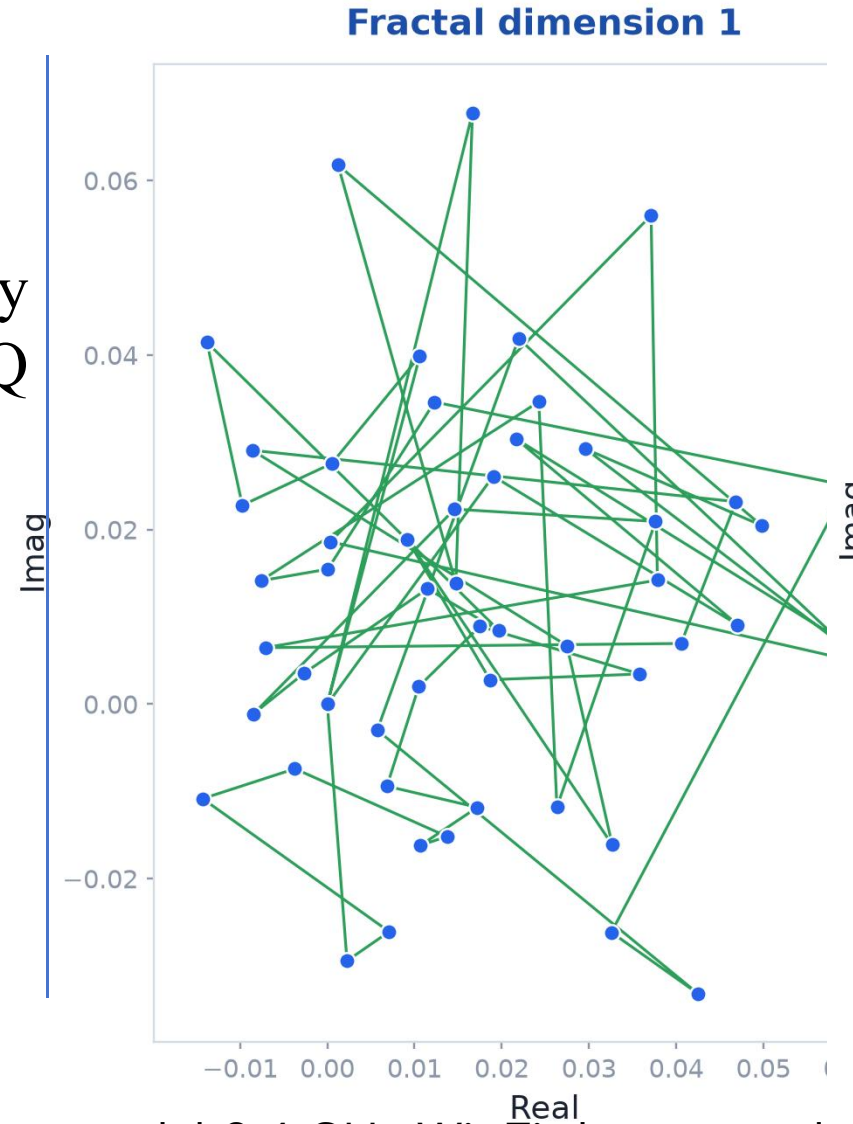
Representative RF Features – Fractal Dimension

Fractal dimension

Quantifies the complexity and self-similarity of the I/Q signal distribution

Interpretation

Different I/Q structures → different fractal dimensions



Ref.:

- SMOFFI: A large-scale same-model 2.4 GHz Wi-Fi dataset and reproducible framework for RF fingerprinting, Computer Networks, Volume 282, 2026.

Representative RF Features – Summary

- **CFO** → frequency offset
- **Phase Error** → phase deviation
- **Magnitude Error** → magnitude deviation
- **I/Q Gain Imbalance** → I/Q energy imbalance
- **Fractal Dimension** → I/Q distribution complexity/self-similarity